ClipSAT : Cambridge & Edexcel Additional Mathematics 0606/4PM1

Unit 6.01: Permutations & Combinations

Name: _____

Q1. Find the value of the integer n ge 3 that satisfies the equation {}^{n}mathrm{P}_{3} = 10 times {}^{n}mathrm{C}_{2}

[ ] A: n = 7
[ ] B: n = 12
[ ] C: n = 5
[ ] D: n = 8

Q2. A committee of 5 members is to be selected from a group consisting of n men and 6 women, where n ge 4 . Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n .

[ ] A: 13
[ ] B: 10
[ ] C: 8
[ ] D: 7

Q3. Six-digit numbers are formed using digits chosen from the set {1, 2, 3, 4, 5, 6, 7, 8}, with no digit repeated in any number. Find the number of different six-digit odd numbers that can be formed such that the digits 2 and 4 are not adjacent.

[ ] A: 8160
[ ] B: 7680
[ ] C: 9120
[ ] D: 2880

Q4. Given that {}^{n}mathrm{P}_{3} = 8,{}^{n}mathrm{C}_{2}, find the value of the integer n , where n ge 3 .

[ ] A: 6
[ ] B: 8
[ ] C: 10
[ ] D: 12

Q5. A group of n students, which includes Ali and Priya, are to be arranged in a straight line for a photograph. The number of different arrangements in which Ali and Priya are not standing next to each other is 3600 . Find the value of the integer n .

[ ] A: 6
[ ] B: 7
[ ] C: 8
[ ] D: 9

Q6. Two parallel lines, L_1 and L_2 , have n and n + 2 distinct points marked on them respectively, where n is an integer with n ge 2 . Triangles are formed by choosing 3 of these points as vertices such that they are not collinear. Quadrilaterals are formed by choosing 4 of these points as vertices such that no three are collinear. Given that the ratio of the number of quadrilaterals to the number of triangles that can be formed is 6 : 5 , find the total number of triangles that can be formed.

[ ] A: 175
[ ] B: 210
[ ] C: 220
[ ] D: 96

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Q7. Given that {}^\ntext{P}_2 = \dbinom{n}{3}, where n is an integer such that n \ge 3, find the value of n.

[ ] A: 8
[ ] B: 5
[ ] C: 3
[ ] D: 6

Q8. A committee of 4 people is to be chosen from 5 men and 4 women. How many committees consist of exactly 2 men and 2 women?

[ ] A: 60
[ ] B: 120
[ ] C: 55
[ ] D: 36